AWS Certified AI Practitioner Part 1

Generative AI & Amazon Bedrock

Amazon Bedrock

Generative AI

- A subset of deep learning designed to generate new data similar to training data
- Works across text, images, audio, code, and video
- Learns patterns rather than simply copying training examples

Foundation Models

- Large, pretrained models trained on broad datasets
- Adaptable to many tasks: generation, summarization, extraction, Q&A, chatbots
- Training from scratch can require enormous compute and cost
- Major model developers include OpenAI, Meta, Amazon, Google, and Anthropic

Large Language Models (LLMs)

- Foundation models specialized for understanding and generating human-like text
- Common tasks: translation, summarization, Q&A, extraction, creation, classification
- Generate responses token-by-token using probabilities
- Same prompt can sometimes produce different outputs

“After the rain, the streets were…”

Amazon Bedrock

- Fully managed AWS service for building and deploying generative AI applications
- Access foundation models through APIs without managing underlying servers
- Data is not used to train underlying foundation models by default
- Capabilities include model access, RAG, Agents, fine-tuning, evaluation, and Guardrails

Choosing a Foundation Model

Consider:

• Performance & capabilities • Cost • Compliance • Customization

• Model size • Inference options • Context window • Latency

• Input/output modalities • Service agreements and constraints

Multimodal models can work with multiple data types such as text, images, audio, and video.

Amazon Titan

- AWS foundation model family supporting generative AI use cases
- Includes text and embedding capabilities
- Model size matters: smaller models can be more cost-effective when performance is sufficient
- Goal: balance capability, performance, latency, and cost

Fine-Tuning

- Adapts a copy of a foundation model using your own training data

- Model weights are updated for a particular task or domain

- Training data follows required formats and is typically stored in Amazon S3, depending on service/model

- Not every foundation model supports fine-tuning

Supervised vs. Reinforcement Fine-Tuning

SUPERVISED

- Uses labeled input-output examples
- Improves performance on specific tasks

REINFORCEMENT

- Uses feedback and a reward function
- Responses receive scores based on desired criteria
- Model learns to maximize reward

Key distinction: labeled examples vs. feedback/rewards.

Model Distillation

- Transfers knowledge from a larger model into a smaller model
- Smaller models can be faster and less expensive
- Can reduce inference cost for a specific use case
- Tradeoff: potentially lower accuracy or fewer capabilities

Fine-Tuning Use Cases

- Specific persona, tone, communication style, or business purpose
- Recent or specialized information
- Exclusive/proprietary data such as customer-service interactions

Important: fine-tuning is not always best; RAG can add information without changing the model.

Part 1 Review

Know the differences:

- Generative AI vs. foundation models vs. LLMs
- What Amazon Bedrock manages
- How to choose a model
- Fine-tuning vs. supervised vs. reinforcement fine-tuning
- Distillation and its performance/cost tradeoff